

Supplementary Material for:
Safe Active Discrimination of Fractional, Delayed, and Finite
Latent Memory in a Strong-Allee Predator–Prey Model

Ibrahim Alraddadi
Department of Mathematics, Faculty of Science
Islamic University of Madinah, Madinah, Saudi Arabia
`ialraddadi@iu.edu.sa`

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This supplement contains the complete proofs, derivations, computational-layer specifications, prospective experimental protocols, and numerical validation material that support the main paper. Numbered references of the form “Theorem 6.5” or “Equation (42)” resolve to the main paper; objects numbered “ Sn ” are local to this supplement. Nothing in this supplement introduces a claim beyond those stated in the main paper.

S1 Theorem index and proof locations

Every theorem of this paper is proved exactly once, at the point of its statement, in the regime (finite-dimensional, infinite-dimensional, linearized, or nonlinear) in which it is used. No theorem carries a second proof elsewhere in the manuscript. The proofs rely on the structural separation results of Section 5 [8, 2, 3, 5, 6, 7, 4, 1]. Tables S1 and S2 record, for each result, its exact statement location and the assumptions under which it holds.

Two editorial rules from the reconstruction are enforced throughout. First, mathematical theorems are separated from computational algorithms (the nested Monte Carlo estimator of Section S5.3, the constrained optimizations of Section S3.3) and from empirical observations (Section 10): an algorithm or a benchmark number is never stated as a theorem, and no theorem proof introduces constraints or variables absent from its statement. Second, each statement declares its regime — finite-dimensional or function-space, linearized or nonlinear — so that linear-Gaussian certificates are never read as nonlinear claims and conversely.

S2 Approximation: omitted proofs and explicit parameter choices

This section contains the complete proof of the constructive L^1 approximation theorem (Theorem 6.3 of the main paper) and the explicit parameter choices achieving a prescribed tolerance.

Proof of Theorem 6.3 (main paper). Split the integral (1) into three regions: $(0, \ell)$, $[\ell, L]$, and (L, ∞) .

Result	Statement	Regime and key assumptions	Location
Thm 5.1	Caputo transfer not rational	fixed models, analytic continuation; $CB \neq 0$	Sec. 5
Thm 5.2	Caputo vs. retarded delay	strictly proper retarded, finite state dimension, single discrete delay, no feedthrough	Sec. 5
Thm 5.4	Neumann remainder	$ q_\alpha(s) \geq 2\ J\ $	Sec. 5
Thm 5.5	Separating frequency	finite collection of fixed analytic transfers; compact band away from poles and branch point	Sec. 5
Thm 6.1	Positive diffusive representation	$0 < \alpha < 1$, exact kernel identity	Sec. 6
Thm 6.2	No uniform approximation at 0	kernel singularity, L^∞ norm	Sec. 6
Thm 6.3	Constructive L^1 approximation	explicit construction, finite horizon	Sec. 6
Thm 6.4	Output approximation	Young's inequality, fixed $u \in L^2$	Sec. 6
Thm 6.5	Finite-horizon barrier	linearized, fixed input	Sec. 6
Thm 6.9	Testing bound under safe excitation	frozen $\mathcal{K}_m$; interval-enclosed $\hat{E}_m^{IA}$; bounded observation operator; outer budget B_{eff} ; common Gaussian covariance	Sec. 6
Thm 6.10	Hierarchy-uniform Strong-Allee budget on a fixed protocol ray	ecological realization via ODE backbone; positive mode measure; mass bound M_T ; exact supremum attained by single-mode kernel	Sec. 6
Lem. 6.12	Pole/branch-cut decomposition of the fractional relaxation kernel	$\alpha\pi/2 < \arg \lambda < \alpha\pi$ (interval-verified per cell)	Sec. 6
Thm 6.13	Certified ecological prey-response approximation ($\hat{E}_m^{\text{state}}$)	exact prey-to-prey G_α ; strictly proper finite-state LTI response class $\mathcal{G}_m$; pole region and L^1 mass cap; interval enclosures	Sec. 6
Cor. 6.14	Prey-response testing bound	bounded prey observation; independently valid outer L^2 input budget; same response hierarchy $\mathcal{G}_m$	Sec. 6
Cor. 6.6	Unrestricted-dimension obstruction	linearized fixed-input scope; see Rem. 6.7	Sec. 6

Table S1: Theorem index, part I: structural separation, finite-horizon approximation, and testing results.

Low-rate tail $(0, \ell)$: integrating the density over time,

$$\int_0^T \int_0^\ell \lambda^{-\alpha} e^{-\lambda t} d\lambda dt \leq T \int_0^\ell \lambda^{-\alpha} d\lambda = \frac{T\ell^{1-\alpha}}{1-\alpha}.$$

High-rate tail (L, ∞) : by Tonelli,

$$\int_0^T \int_L^\infty \lambda^{-\alpha} e^{-\lambda t} d\lambda dt = \int_L^\infty \lambda^{-\alpha-1} (1 - e^{-\lambda T}) d\lambda \leq \frac{L^{-\alpha}}{\alpha}.$$

Middle interval $[\ell, L]$: since $\lambda^{-\alpha} e^{-\lambda t}$ is decreasing in λ for fixed t , the gap between left and right Riemann sums is at most

$$h \left(\ell^{-\alpha} e^{-\ell t} - L^{-\alpha} e^{-Lt} \right) \leq h\ell^{-\alpha}.$$

Integrate over $t \in [0, T]$, multiply by c_α , and sum the three contributions. $\square$

Result	Statement	Regime and key assumptions	Location
Thm 7.1	Energy-optimal pairwise input	linear-Gaussian; attainment via compactness or finite discretization	Sec. 7
Cor. 7.3	Optimal single frequency	fixed simple models	Sec. 7
Thm 7.5	Finite multisine support	P fixed continuous scores on compact Ω ; Carathéodory	Sec. 7
Thm 7.7	Optimal sampling schedule	independent measurements, no coupling constraints	Sec. 7
Thm 8.1	Safe informative perturbation	local well-posedness, continuous solution maps, interior margin	Sec. 8
Lem. 8.3	Linear safety envelope	stable linear Caputo system, amplitude bound	Sec. 8
Thm 8.4	Strict inward-pointing rectangle	$z(0) \in \text{int } \mathcal{R}$, uniform margin $\eta > 0$, Caputo endpoint identity	Sec. 8
Thm 8.6	Safety alone does not bound input energy	$h_x \in L^1$; one-sided floor; Riemann–Lebesgue	Sec. 8
Thm 9.1	Nonlinear posterior update	Gaussian noise, possibly nonlinear mean map; linear-Gaussian reduction marked	Sec. 9
Thm 9.2	Sequential information identity	expected-KL form; see Rem. S5.2, S5.3	Sec. 9

Table S2: Theorem index, part II: design, safety, and Bayesian results.

S2.1 Explicit parameter choice for tolerance epsilon

For a given tolerance ε , it suffices to choose

$$\ell = \left(\frac{\varepsilon(1-\alpha)}{3c_\alpha T} \right)^{1/(1-\alpha)}, \quad (\text{S1})$$

$$L = \left(\frac{3c_\alpha}{\alpha\varepsilon} \right)^{1/\alpha}, \quad (\text{S2})$$

$$h \leq \frac{\varepsilon\ell^\alpha}{3c_\alpha T}, \quad m \geq \frac{L-\ell}{h}. \quad (\text{S3})$$

This construction is conservative, not complexity-optimal; it is sufficient for an existence theorem. In practice, faster convergence can be achieved with optimized Gauss–Laguerre or adaptive quadrature, but the construction above guarantees polynomial scaling.

Proof of Theorem 6.9 (testing bound under bounded safe excitation).

By the definition of $\hat{E}_m^{\text{IA}}$ there is an explicit padded competitor $\tilde{k}_m \in \mathcal{K}_m$ with $\|k_\alpha - \tilde{k}_m\|_1 \leq \hat{E}_m^{\text{IA}}$. For every $u \in \mathcal{U}_m$, Young’s inequality gives

$$\|(k_\alpha - \tilde{k}_m) * u\|_2 \leq \hat{E}_m^{\text{IA}} B_{\text{eff}}.$$

Hence the Gaussian KL divergence of this fixed two-point subproblem satisfies

$$D_{\text{KL}}(P_{C_\alpha}^u \| P_{\tilde{k}_m}^u) = \frac{1}{2} \|R^{-1/2} S((k_\alpha - \tilde{k}_m) * u)\|_2^2 \leq U_m.$$

The composite minimax risk is at least the Bayes risk of this particular two-point subproblem. Pinsker’s inequality and Bretagnolle–Huber therefore give the first two lower bounds [9]; for common-covariance Gaussians the two-point Bayes error is exactly $\Phi(-\sqrt{D_{\text{KL}}/2})$, yielding the third because this function decreases with KL. Since the KL ceiling is uniform in $u \in \mathcal{U}_m$, taking the outer infimum over experiments preserves the bound.

Proof of Theorem 6.10 (hierarchy-uniform Strong-Allee budget). Upper bound: by linearity in the positive mode measure,

$$d(k) \leq \sum_j c_j F(\lambda_j) = \sum_j [c_j w(\lambda_j)] \frac{F(\lambda_j)}{w(\lambda_j)} \leq M_T \max_\lambda \frac{F(\lambda)}{w(\lambda)},$$

because $\sum_j c_j w(\lambda_j) = \|k\|_{L^1(0,T)} \leq M_T$. Attainment: the single-mode kernel k^* has mass exactly M_T , one rate in the window, and achieves $d(k^*) = M_T F(\lambda^*)/w(\lambda^*)$, so the supremum is a maximum and the stated value is exact. One-sided safety $x^* + x_k(t) \geq A + \eta$ for all $k \in \mathcal{K}_m$ forces $c d(k) \leq \rho_\eta$ for every k , hence $c \leq \rho_\eta/d_{\text{rob}}(\mathcal{K}_m, u_0)$, and multiplying by $\|u_0\|_2 = 1$ gives the bound.

Proof of Lemma 6.12 (pole/branch-cut decomposition). Deform the Bromwich contour for $(s^\alpha - \lambda)^{-1}$ onto a Hankel contour about the negative real axis, using the principal branch of s^α . The sector condition gives exactly one principal-sheet zero $s_0 = \lambda^{1/\alpha}$, with residue $(s_0/(\alpha\lambda))e^{s_0 t}$. The jump across $s = re^{\pm i\pi}$ is

$$\frac{1}{r^\alpha e^{-i\alpha\pi} - \lambda} - \frac{1}{r^\alpha e^{i\alpha\pi} - \lambda} = \frac{2ir^\alpha \sin(\alpha\pi)}{r^{2\alpha} - 2\lambda r^\alpha \cos(\alpha\pi) + \lambda^2},$$

which yields the stated branch integral. The strict upper sector bound keeps the denominator nonzero for $r > 0$; the small- and large-arc contributions vanish in the standard Hankel deformation for $t > 0$.

S3 Design: omitted proofs and practical layers

This section contains the complete proofs of the energy-optimal input theorem (Theorem 7.1) and the finite multisine sufficiency theorem (Theorem 7.5), followed by the three practical design layers summarized in the main paper: spacing- and cost-constrained observation design with channel selection, the contrast between discrimination and parameter-estimation designs, and constrained implementation.

Proof of Theorem 7.1. Substituting (17) and (19) into (18) yields

$$D_{12}(u) = \frac{1}{2} \langle u, Qu \rangle.$$

The constrained supremum $\sup_{\|u\|^2 \leq E} \langle u, Qu \rangle$ is a Rayleigh quotient problem. Because $Q \succeq 0$, the Rayleigh–Ritz theorem gives

$$\sup_{\|u\|^2 \leq E} \langle u, Qu \rangle = E \lambda_{\max}(Q),$$

with equality at any unit principal eigenvector whenever the top spectral value is an eigenvalue. This is automatic in the finite-dimensional discretization used for computation. In the infinite-dimensional input-function setting, attainment additionally requires the whitened difference operator $A = R^{-1/2} \Delta H : \mathcal{U} \rightarrow \mathcal{Y}$ to be compact between Hilbert spaces: then $Q = A^* A$ is compact, self-adjoint, and positive, its spectral radius is an eigenvalue, and a principal eigenfunction exists. Rescaling by $\sqrt{E}$ saturates the energy constraint and attains the bound. Without compactness or an explicit finite discretization, the supremum value (20) still holds but need not be attained, and no eigenfunction maximizer should be claimed.

Proof of Theorem 7.5. *Existence.* The feasible set of nonnegative measures with total mass at most E is weakly compact, and the constraints define a closed affine slice, so the supremum is attained.

Finite support. Take any nonzero feasible ξ and normalize it to a probability measure $\tilde{\xi} = \xi/\xi(\Omega)$. The vector of scaled pairwise scores

$$\bar{w}(\tilde{\xi}) := \left(\int_{\Omega} w_1 d\tilde{\xi}, \dots, \int_{\Omega} w_P d\tilde{\xi} \right) \in \mathbb{R}^P$$

lies in the convex hull of the curve

$$\mathcal{W} := \{(w_1(\omega), \dots, w_P(\omega)) : \omega \in \Omega\} \subset \mathbb{R}^P.$$

By Carathéodory's theorem in $\mathbb{R}^P$, every point in $\text{conv}(\mathcal{W})$ admits a convex representation using at most $P + 1$ points of $\mathcal{W}$. Therefore $\tilde{\xi}$ (and hence ξ) may be replaced by a measure supported on at most $P + 1$ frequencies without changing any of the P integrated scores. Rescaling back to total mass E preserves feasibility and the objective value t .

S3.1 Practical observation design: spacing, cost, and channel selection

The top- n rule of Theorem 7.7 is exact but idealized. Real experiments impose three corrections.

Minimum-spacing constraint

For a single observation channel with candidate times $t_1 < \dots < t_N$ and a minimum index spacing L between consecutive observations, the constrained exact algorithm is the dynamic program

$$V(k, j) = \max\{V(k - 1, j), s(t_k) + V(k - L, j - 1)\}, \quad (\text{S4})$$

where $V(k, j)$ is the best achievable total score using j observations among the first k candidate times. This solves the spacing-constrained pairwise schedule in $O(Nn)$ operations.

Variable costs and budget

When each candidate (t, o) carries a measurement cost $c(t, o)$ and the total sampling budget is B , select binary variables $z_{t,o} \in \{0, 1\}$ and solve the 0–1 knapsack

$$\max \sum_{t,o} s_{12}(t, o) z_{t,o} \quad \text{s.t.} \quad \sum_{t,o} c(t, o) z_{t,o} \leq B, \quad z_{t,o} \in \{0, 1\}. \quad (\text{S5})$$

For P model pairs, introduce a worst-case floor q and the P constraints $\sum_{t,o} s_p(t, o) z_{t,o} \geq q$ for every pair p , yielding a maximin mixed-integer linear program. This is the observation-side analogue of (25).

Channel selection from the strong-Allee linearization

The observability analysis of Section 4 establishes three structural facts that frame channel choice: (i) prey-only observation is structurally sufficient; (ii) predator-only observation is structurally sufficient; (iii) observing both species is *not* mathematically mandatory for state observability.

However, the direct-channel asymptotics of Theorem 5.1 refine this picture for discrimination:

- Perturb prey and measure prey to expose the leading $s^{-\alpha}$ signature directly.
- Perturb predator and measure predator for the same reason via the predator self-channel.
- Use the cross-species channel (prey input, predator output, or vice versa) to estimate the interaction parameters c, d . Cross channels have $CB = 0$, so their leading asymptotic decay is $s^{-2\alpha}$ (or lower) with a correspondingly different high-frequency phase; the collocated phase gap $(1 - \alpha)\pi/2$ of (15) must not be attributed to them.

The conclusion is that the preferred practical design is *not* “both species at every time”. It is:

1. high-rate collocated measurement around perturbation/switching times, where the principal eigenfunction of Theorem 7.1 concentrates energy;
2. lower-rate cross-species measurement across the recovery interval, to resolve interaction parameters;
3. environmental covariate measurement only when the Popov–Belevitch–Hautus latent-mode condition is weak or prior uncertainty makes latent forcing competitive.

S3.2 Why discrimination and parameter-estimation designs differ

It remains to state precisely why the schedules of Theorems 7.1–7.7 are not interchangeable with designs aimed at parameter estimation. For a parameter vector ψ , local parameter estimation uses the sensitivity operator

$$S_\psi = \frac{\partial H(\psi)}{\partial \psi}, \quad (\text{S6})$$

evaluated at a nominal model. The associated Fisher information is

$$F(u) = S_\psi(u)^* R^{-1} S_\psi(u), \quad (\text{S7})$$

whereas the discrimination objective uses the inter-model operator

$$Q_{12} = \Delta H^* R^{-1} \Delta H. \quad (\text{S8})$$

These are different operators. A design that makes F well-conditioned can simultaneously leave $\Delta H u$ small, because a parameter-tunable latent model tracks the fractional response at the nominal operating point — exactly the failure mode classified as an adaptive-model risk in Section 5. Conversely, a design that maximizes pairwise divergence maximin-robustly (Theorem 7.5) need not render every

parameter of either model individually identifiable. D-optimal parameter design maximizes $\log \det F$, an objective that rewards complementary sensitivity directions; the top- n separation theorem (Theorem 7.7) does *not* apply to D-optimal schedules because the determinant is not additive.

This operational distinction is one of the central findings of the paper: *good inputs for parameter estimation are not automatically good inputs for model discrimination*, and vice versa. The experiment protocol developed in later sections therefore separates the two objectives and runs a two-stage design: a discrimination stage that maximizes the worst-case pairwise divergence via Theorems 7.1–7.5, followed by an estimation stage conditioned on the selected model, planned via (S7) with the discrimination data as prior.

S3.3 Implementation under additional constraints

The unconstrained eigenvector solution of Theorem 7.1 is the benchmark and the upper bound for any constrained implementation. When box, rate, or cumulative-impact constraints apply, the design becomes

$$\max_u \frac{1}{2} u^* Q u \quad \text{s.t.} \quad |u_k| \leq u_{\max}, \quad |u_{k+1} - u_k| \leq r_{\max}, \quad \sum_k |u_k| \Delta t \leq B, \quad (\text{S9})$$

which is generally a nonconvex quadratic maximization. Reliable solution methods include:

- direct transcription with multiple random starts, exploiting the structure of Q ;
- semidefinite relaxation with randomized rounding, certifying the eigenvector optimum as the outer bound;
- dynamic programming for low-dimensional discrete input alphabets;
- Bayesian optimization when the forward ecological simulator is expensive and gradients of H_i are unavailable.

In every case the unconstrained u^* of (21) remains the natural warm start and the ceiling against which realized performance is scored. The finite-support guarantee of Theorem 7.5 further reduces the search space: once the $P + 1$ optimal tones are identified, the constrained problem reduces to choosing their complex amplitudes subject to the physical constraints, a much smaller nonconvex program than full-waveform input optimization.

The combined input-plus-observation design of this section — Theorems 7.1, 7.5, and 7.7 together with the practical refinements of Section S3.1 — is what converts the structural separation of Section 5 and the complexity-bounded approximation barrier of Section 6 into a deployable, computable discrimination experiment. The remaining sections address the safety of that experiment (Section 8), its Bayesian extension beyond the linear-Gaussian regime (Section 9), and a reproducible benchmark (Section 10) that implements the constructions above on the strong-Allee predator–prey model.

S4 Safety: omitted proofs, face inequalities, and practical consequences

This section contains the complete proofs of the linear safety envelope (Lemma 8.3) and the fractional strict inward-pointing rectangle theorem (Theorem 8.4), the derivation of the four Strong-Allee face inequalities used by the input-envelope procedure of the main paper, and the practical consequences of operating near the Allee threshold.

Proof of the linear safety envelope (Lemma 8.3). From (34),

$$\|\xi(t)\| \leq \int_0^t \|H_\alpha(t-s)\| \|u(s)\| ds \leq \|u\|_{L^\infty} \int_0^t \|H_\alpha(\tau)\| d\tau \leq \|u\|_{L^\infty} \Gamma_T.$$

Inserting the bound (36) yields $\|\xi(t)\| \leq \rho$ for all $t \in [0, T]$.

Proof of Theorem 8.4. Suppose for contradiction that some component first contacts a face of $\mathcal{R}$. Consider the lower face: there is a first time $t_* > 0$ and a component i with $z_i(t_*) = \ell_i$ and $z_i(t) > \ell_i$ for all $t \in [0, t_*)$. Since $z(0) \in \text{int } \mathcal{R}$, we have $z_i(0) > \ell_i$. On $[0, t_*]$, the component z_i attains its minimum at the right endpoint t_* . We use the following Caputo endpoint identity, valid for any absolutely continuous w on $[0, t_*]$:

$${}^C D_t^\alpha w(t_*) = \frac{\alpha}{\Gamma(1-\alpha)} \int_0^{t_*} \frac{w(t_*) - w(s)}{(t_* - s)^{1+\alpha}} ds,$$

obtained by integrating the definition ${}^C D_t^\alpha w(t) = \frac{1}{\Gamma(1-\alpha)} \int_0^t (t-s)^{-\alpha} w'(s) ds$ by parts; the integrated term vanishes because $(t_* - s)^{-\alpha} (w(t_*) - w(s)) \rightarrow 0$ at both endpoints. Applying this with $w = z_i$, every integrand value satisfies $z_i(t_*) - z_i(s) = \ell_i - z_i(s) \leq 0$, with strict inequality on a set of positive measure (since $z_i(0) > \ell_i$ and z_i is continuous). Hence

$${}^C D_t^\alpha z_i(t_*) < 0.$$

But the dynamics and the strict inward-pointing condition (39) give ${}^C D_t^\alpha z_i(t_*) = F_i(t_*, z(t_*)) \geq \eta > 0$, a contradiction. The upper-face case is identical with signs reversed, using that z_i attains its maximum at t_* . Hence no component can have a first contact with a face, and the solution stays in $\text{int } \mathcal{R}$.

S4.1 Strong-Allee boundary inequalities

Applying Theorem 8.4 to the controlled strong-Allee predator-prey system from Section 2 with additive controls u_x, u_y and safety rectangle

$$\mathcal{R} = [x_L, x_U] \times [y_L, y_U], \tag{S10}$$

the four boundary conditions become explicit deterministic inequalities. For brevity we present the four sufficient conditions; each may be tightened by worst-case analysis over parametric uncertainty using interval arithmetic.

Lower prey face

At the lower prey boundary $x = x_L$, the vector field component is

$$F_x = rx_L \left(1 - \frac{x_L}{K}\right) \left(\frac{x_L}{A} - 1\right) - \frac{ax_L y}{1 + hx_L} + b_x u_x.$$

Taking the worst-case prey depletion at the maximum plausible predator abundance y_U , the sufficient inward-pointing condition is

$$\boxed{rx_L \left(1 - \frac{x_L}{K}\right) \left(\frac{x_L}{A} - 1\right) - \frac{ax_L y_U}{1 + hx_L} + b_x u_x^{\min} \geq 0}. \quad (\text{S11})$$

Upper prey face

At the upper prey boundary $x = x_U$ with the minimum predator y_L ,

$$\boxed{rx_U \left(1 - \frac{x_U}{K}\right) \left(\frac{x_U}{A} - 1\right) - \frac{ax_U y_L}{1 + hx_U} + b_x u_x^{\max} \leq 0}. \quad (\text{S12})$$

Lower predator face

At the lower predator boundary $y = y_L$,

$$\boxed{y_L \left(\frac{eax}{1 + hx} - m\right) + b_y u_y^{\min} \geq 0, \quad \forall x \in [x_L, x_U]}. \quad (\text{S13})$$

The parenthesis is the functional response gradient. The inequality must hold for every prey value in of x in the box, which can be verified by evaluating its minimum over $[x_L, x_U]$ (a one-dimensional check for this monotonic Holling type II response).

Upper predator face

At the upper predator boundary $y = y_U$,

$$\boxed{y_U \left(\frac{eax}{1 + hx} - m\right) + b_y u_y^{\max} \leq 0, \quad \forall x \in [x_L, x_U]}. \quad (\text{S14})$$

The inequalities (S11)–(S14) are sufficient, not necessary: a vector field can fail the sign condition at a boundary without the trajectory reaching it. They are adequate for design, and numerical verification prior to each experiment is straightforward. For the locked parameters of Section 2 — $r = 3/2, K = 1, a = 1, h = 1/2, e = 4/5, m = 2/5$ — the left-hand sides are rational functions of A .

S4.2 Practical consequences of the Allee threshold

The Allee threshold A introduces two specific warnings that shape any implementable protocol.

Negative prey pulses. Removing prey from a population near the Allee threshold is the most informative perturbation under some model pairs because it probes the nonlinear prey growth function directly and can disentangle the sign of the growth rate. However, a negative prey pulse directly pushes the population toward A . The inequalities (S11)–(S14) require that even at maximum removal, the net prey impulse does not cross the lower bound $x_L > A$. In practice, this means:

- Reserve a certified margin $\|x(t) - A\| \geq \Delta_{\min} > 0$ at all times.
- Pre-emption removal by positive predator pulses when the lower-face condition fails.
- Use positive prey nutrients or predator removal perturbations for the harvesting-like signal whenever it is scientifically equivalent.
- Adaptive termination: stop the trial automatically if a posterior predictive probability of crossing x_L exceeds a user-specified safety level β (e.g., 5%).

Cumulative impact. The linear safety bound of Lemma 8.3 is an instantaneous envelope that does not account for the accumulated distortion of repeated perturbation–recovery cycles. For protocols with many sequential pulses, add the cumulative-integral constraint

$$\sum_k \|u_k\| \Delta t_k \leq C_{\max}, \quad (\text{S15})$$

where the $C_{\max}$ is calibrated by numerical simulation of the worst-case admissible trajectory. This guarantee is the robust analogue of the linear impulse bound for repeated perturbation designs.

The safety architecture of this section — existence guarantee (Theorem 8.1) plus invariance certificate (Theorem 8.4) — formally closes the loop between the information-theoretic optima of Section 7 and the ecological admissible parameter described in Section 2. The practical envelope procedure above and its algorithmic counterpart will be exercised numerically in Section 10, and its Bayesian extension (for parameter uncertainty on parameters including the Allee threshold itself) is developed in Section 9.

Proof of Theorem 8.1 (safe and informative perturbation exists). By continuity of the solution maps and the positive interior margin ρ , there exists $\varepsilon_s > 0$ such that the state remains in $\mathcal{S}$ for both models whenever $\|\varepsilon\| < \varepsilon_s$. The output difference expands as

$$\mu_1(u_\varepsilon) - \mu_2(u_\varepsilon) = \varepsilon(H_1 - H_2)v + o(\varepsilon). \quad (\text{S16})$$

Since the leading vector is nonzero by condition (ii), the difference is nonzero for sufficiently small positive ε . Under the nonsingular Gaussian noise model of Section 7, the Kullback–Leibler divergence is a positive quadratic form in $\Delta\mu$ (Equation 18). Choose ε_0 below both ε_s and the threshold at which the $o(\varepsilon)$ term dominates the linear term. Then every $0 < \varepsilon < \varepsilon_0$ is simultaneously safe and informative.

S5 The Bayesian sequential layer: proofs, estimator, schedule, and robustness axes

This section completes the Bayesian layer of the main paper: the joint model-plus-parameter utility, the complete proofs of Theorems 9.1 and 9.2 with their scope-limiting remarks, the nested Monte Carlo estimator and its variance-reduction hierarchy, the adaptive schedule with stopping rules, and the eight mandatory robustness axes. The layer is specified, not executed, in this work.

S5.1 Joint model-plus-parameter utility

If the purpose includes parameter recovery — that is, if the goal is to refine the within-model parameters and not only to separate models — the joint utility replaces the model marginal by the joint posterior:

$$U_{\text{joint}}(d) = I((M, \vartheta_M); Y \mid d). \quad (\text{S17})$$

The objective (S17) can favor designs that are locally informative for within-model parameter estimation but not robust for separating models whose parametric families overlap; the paper therefore reports model-only (42) and joint (S17) utilities separately, and never as a single commensurable scalar.

S5.2 Complete proofs and remarks

Proof of Theorem 9.1. Bayes’ rule applied to the observation model (45) gives, for a single step, $p(\vartheta_M \mid Y_k, M, d_k) \propto p(\vartheta_M \mid M) \mathcal{N}(Y_k; \mu_M(\vartheta_M; d_k), R)$. Independence of the additive noise across steps together with the forward factorization $\prod p(Y_k \mid \vartheta_M, M, d_k)$ yields the product form (46); the model-level update (47) follows by integrating out ϑ_M at each model and reweighting through Bayes. In the linear-Gaussian conjugate case, μ_M is affine in ϑ_M and the prior is Gaussian, so the product likelihood is Gaussian in the parameter and the within-model marginal integrates exactly to a Gaussian predictive density with covariance $H_M P_M H_M^\top + R$; the pairwise separation of these predictive means is governed by the operator $Q = \Delta H^* R^{-1} \Delta H$ of Section 7. Outside that conjugate case the product likelihood is generically non-Gaussian in ϑ_M — the Gaussianity of (45) holds only conditionally on the parameter — and the integral in (47) must be evaluated numerically.

Remark S5.1. Theorem 9.1 mediates the boundary between the analytical layer of Section 7 and the computational layer of the present section. The closed-form optimal input u^* of Theorem 7.1 is recovered exactly when the map μ_M is linear in the parameter; in any other regime u^* is the screening benchmark — the upper bound that the nested estimator of Section S5.3 is calibrated against — but not the implemented design.

Proof of Theorem 9.2. Write $\mathcal{D}_n = (Y_{1:n}, d_{1:n})$. Since d_{n+1} is a function of $\mathcal{D}_n$, conditioning on d_{n+1} adds no information about M beyond $\mathcal{D}_n$. The chain rule for mutual information gives

$$I(M; Y_{n+1} \mid \mathcal{D}_n, d_{n+1}) = H(M \mid \mathcal{D}_n) - H(M \mid \mathcal{D}_n, Y_{n+1}, d_{n+1}).$$

Expanding the conditional mutual information as an expectation of log-ratios yields (50): for each realized Y_{n+1} the realized posterior $p(M \mid \mathcal{D}_n, Y_{n+1}, d_{n+1})$ is compared, in KL, against the current posterior $p(M \mid \mathcal{D}_n)$, and the comparison is averaged under the prior predictive distribution of Y_{n+1} . The upper bound (51) follows because conditional entropy is nonnegative, so $I(M; Y_{n+1} \mid \mathcal{D}_n, d_{n+1}) \leq H(M \mid \mathcal{D}_n) = H(M \mid Y_{1:n})$; the lower bound is the nonnegativity of mutual information. Equality in the upper bound holds exactly when $H(M \mid \mathcal{D}_n, Y_{n+1}, d_{n+1}) = 0$, i.e. when a single experiment resolves the model identity almost surely.

Remark S5.2 (Expected KL, not KL of the expected posterior). The identity (50) is an *expectation of divergences between realized posteriors* and the current posterior. It must not be replaced by the divergence between the expected future posterior $\tilde{q}(M) = \mathbb{E}_{Y|\mathcal{D}_n, d}[p(M | \mathcal{D}_n, Y, d)]$ and the current posterior: by the tower property, $\tilde{q} = p(M | \mathcal{D}_n)$ identically, so $D_{\text{KL}}(\tilde{q} \| p(M | \mathcal{D}_n)) = 0$ for every design and cannot serve as an acquisition function. The averaging in (50) is over the convex function $p \mapsto D_{\text{KL}}(p \| p(M | \mathcal{D}_n))$, and it is precisely this convexity that makes the quantity positive whenever data can be informative.

Remark S5.3 (Status of greedy scheduling). Maximizing (50) at every step is the natural myopic policy induced by the bound, and this paper adopts it as the definition of the adaptive schedule (S18). Greedy one-step mutual-information maximization does *not*, by itself, minimize the expected number of experiments needed to drive the posterior past a confidence threshold: such a guarantee requires additional structure (for example adaptive submodularity of the information functional under a declared stopping loss, or a full dynamic-programming solution of the sequential problem). No such result is claimed here; a certified non-myopic schedule is left to future work.

S5.3 Nested Monte Carlo estimator and variance reduction

The objective (42) and its sequential variant (50) have no closed-form evaluation outside the linear-Gaussian regime of Section 7. A direct nested Monte Carlo estimator proceeds in five steps:

1. sample models $M^{(r)} \sim p(M)$ from the current model posterior;
2. sample parameters $\vartheta^{(r)} \sim p(\vartheta | M^{(r)})$;
3. simulate future data $Y^{(r)} \sim p(Y | M^{(r)}, \vartheta^{(r)}, d)$ via the nonlinear Caputo solver of Section 2;
4. estimate each model evidence $p(Y^{(r)} | M_j, d)$ by an inner parameter sample for every candidate model M_j ;
5. average the log-ratio in Equation (42) over r .

Common random numbers are shared across candidate designs to reduce the variance of the design optimizer: only differences in expected utility are then decision-relevant, and correlated noise cancels.

Because nested Monte Carlo is expensive and a single utility evaluation can require an inner ensemble, a strict variance-reduction hierarchy is enforced. Each higher tier is invoked only after the previous tier has been calibrated against a tractable case:

1. *exact linear-Gaussian utility* of Section 7 for screening and unit tests;
2. *Laplace approximation* around posterior modes for evidence and reduced-variance estimators;
3. *importance sampling / sequential Monte Carlo* evidence for posterior multimodality;
4. *neural ratio or mutual-information lower bounds*, admitted only after calibration against the analytic linear-Gaussian case of Tier 1.

The hierarchy is a contract, not a stylistic preference: an estimate at Tier k is admissible only if it recovers Tier $k - 1$ within tolerance on a benchmark where the lower tier is exact. The nesting failure modes — sign bias in the inner evidence at low sample counts, tail sensitivity of the nested log-ratio, and the covariance-smoothing artefact of Laplace approximations at multimodal parameter sets — are diagnosed against Tier 1 before the estimator is used in a live sequential schedule.

S5.4 Adaptive schedule

After each data batch $Y_{1:n}$, the model and parameter posteriors update via Equation (46), and the next design is selected by

$$d_{n+1} = \arg \max_{d \in \mathcal{D}_{\text{safe}}(Y_{1:n})} I(M; Y_{n+1} \mid Y_{1:n}, d), \quad (\text{S18})$$

subject to posterior safety via the chance constraint (48) (or, where required, the robust constraint (49)). The safe set $\mathcal{D}_{\text{safe}}(Y_{1:n})$ incorporates the rectangle inequalities (S11)–(S14) of Section 8 evaluated over the current credible set of parameters — in particular over the Allee threshold A , which enters as a latent variable when it is unknown. The safe input u_{safe} of Section 8 plays the role of a fixed benchmark inside this optimization: the unconstrained optimal input u^* of Theorem 7.1 is used as the screening estimate and the safe projection $\text{sat}_{\rho/\Gamma_T}(u^*)$ is a feasible warm start.

The adaptive schedule terminates when any of the following conditions is met:

- the posterior model probability exceeds a predeclared threshold π_{stop} (e.g. $\pi_{\text{stop}} = 0.95$);
- the expected value of an additional experiment falls below its execution cost, i.e. $I(M; Y_{n+1} \mid Y_{1:n}, d_{n+1}) < c_{\text{exp}}$ relative to a willingness-to-pay;
- no safe design attains the minimum information target D_{min} inside $\mathcal{D}_{\text{safe}}(Y_{1:n})$;
- the model set fails posterior predictive checks, signalling out-of-class dynamics and requiring amplification of the model set rather than further experimentation.

The first two conditions are ordinary stopping rules; the third is the Bayesian counterpart of the safety-for-feasibility trade-off already encountered in Theorem 8.1; the fourth is diagnostic — it triggers the robustness axes of Section S5.5 rather than any further design iteration.

S5.5 Mandatory robustness axes

Every reported “optimal” Bayesian design is re-evaluated, not against a single nominal parameter configuration, but against the following axes of the hierarchical specification. The assessment is mandatory: a design that fails any axis is reported as failing rather than only as nominal-optimal.

- *Prior perturbations* for α , τ , m , and the latent class rates, sampled from a dispersion around each prior;
- *Process versus observation noise*: the balance between the unresolved environmental stochasticity and the measurement noise R is varied over the admissible range;

- *Memory initialization*: a standard Caputo initial-value problem starting at $t = 0$ is determined by point initial data at the lower terminal, not by a DDE-like prehistory function; prior environmental state enters only through the choice of a memory-initialization model (for example an initialized fractional derivative or an exponential-sum state extension), and that model is perturbed within its credible band;
- *Intervention-amplitude error*: the realised actuator amplitude deviates from the commanded u^* within its actuator tolerance;
- *Missing samples*: observation times are dropped uniformly and adversarially to test the schedule’s robustness to sampling dropout;
- *Solver and exponential-sum truncation error*: the Caputo solver and the exponential-sum approximation of Section 6 are run at multiple grid resolutions to bound the truncation bias;
- *Out-of-model functional responses*: the Holling type II response of Section 2 is replaced by a Beddington–DeAngelis and a ratio-dependent response in the data-generating model, and the design is judged on whether the true model remains rejected under each substrate;
- *Correlated environmental forcing*: the white observation noise R is replaced by autocorrelated coloured noise and the utility estimates are recomputed under the generalized likelihood.

The eight axes are reported as heatmaps over the design space, not as pointwise maxima, because the Bayesian optimum is itself a saddle over the perturbation envelope rather than a fixed point.

Theorems 8.1–8.4 of Section 8 and Theorems 9.1–9.2 of this section together draw the full loop: the linear-Gaussian discrimination certificates of Section 7 guarantee that informative safe perturbations exist; the safety invariance theorem of Section 8 guarantees positive invariance only for rectangles whose strict inward-pointing face conditions are verified; the adaptive Bayesian schedule of Equation (S18) chooses the next experiment under posterior safety and terminates at a predeclared confidence threshold; the robustness axes of Section S5.5 re-evaluate every reported design against the known axes of model, noise, and numerical departure. The numerical exercise of this loop, on the strong-Allee benchmark of Section 2, is the subject of Section 10.

S6 Prospective experimental designs A–E and the minimum simulation matrix

This section specifies in full the five prospective microcosm/digital-twin designs summarized in the main paper, and the minimum simulation matrix for prospective validation.

S6.1 Design A — stability-verdict experiment at the certified baseline

Parameters

$$A = 0.3, \quad \alpha = 0.9, \quad \alpha^*(0.3) \approx 0.969012276.$$

The Caputo model is locally stable, whereas the integer-order model with the same Jacobian is unstable.

Intervention Apply a small collocated prey or predator pulse, with amplitude selected by the safety bound of Section 8 (Equation (36)).

Measurements

- high-rate measurement of the perturbed species immediately after the pulse;
- both species during the recovery window;
- enough duration to distinguish algebraic decay from growing/undamped oscillation.

Utility The models differ qualitatively in local stability. This can produce high information but also creates the greatest risk that the unstable alternative leaves the local regime. Use it as a stress-test design, not the sole benchmark.

S6.2 Design B — fair stable frequency-response benchmark

Choose

$$A = \frac{1}{4}.$$

Then

$$J = \begin{pmatrix} -1/8 & -1/2 \\ 1/2 & 0 \end{pmatrix},$$

and the ODE and Caputo alternatives are locally stable.

For $\tau_0 = 1$, $\alpha = 0.9$, and the same J , direct numerical maximization of

$$|G_{0.9}(i\omega) - G_1(i\omega)|^2$$

for each single input/output channel gives a broad optimum near

$$\boxed{\omega\tau_0 \approx 0.49.}$$

This value is a worked numerical result, not a universal constant. It should be recomputed for every parameter posterior and actuator band.

Recommended waveform Use a multisine containing:

- a component near $0.5/\tau_0$;
- a lower-frequency component probing static/recovery behavior;
- a higher-frequency component probing the fractional $-\alpha\pi/2$ phase and $\omega^{-\alpha}$ decay.

The exact frequencies and weights should solve the maximin spectral program (25).

S6.3 Design C — repeated-pulse memory-shape experiment

Let pulse times be logarithmically spaced:

$$t_k = t_0 q^k, \quad q > 1.$$

This probes the kernel over multiple decades. Optimize q , amplitudes, and signs rather than selecting them heuristically.

Rationale A single recovery curve can be matched by a small latent model. Repeated pulses at noncommensurate spacings test superposition across several memory ages and make a low-dimensional exponential approximation less flexible.

S6.4 Design D — latent-environment disambiguation

Use the augmented model

$$\dot{\xi} = J\xi + gz + Bu, \quad \dot{z} = -\lambda z + \gamma u.$$

Calculate

$$\eta_C(\lambda) = |C(J + \lambda I)^{-1}g|.$$

- If η_C is large, the latent mode is visible in population measurements.
- If η_C is small, measure an environmental proxy directly or change the output channel.

This turns “measure environmental covariates” into a quantitative channel-selection rule.

S6.5 Design E — safe adaptive protocol near the Allee threshold

1. choose $x_L > A$ and a safety probability $1 - \delta_s$;
2. begin with a small positive or predator-channel pulse;
3. update model and parameter posteriors;
4. solve the next-step information problem under the chance constraint;
5. prohibit negative prey pulses unless the robust lower-face inequality remains satisfied;
6. stop if no safe design reaches the minimum expected KL/MI.

S6.6 Minimum simulation matrix

Factor	Levels
True model	ODE, Caputo, DDE, latent-1, latent-3
A	0.25, 0.30, 0.40
α	0.70, 0.85, 0.95
observation	prey, predator, both, both+environment
input	pulse, optimized sinusoid, robust multisine, adaptive
SNR	high, medium, low
horizon	short, medium, long
latent dimension cap	1, 3, 6

Table S3: Minimum simulation matrix for prospective microcosm validation.

The latent-dimension cap must be explicit because discrimination against unrestricted m is ill-posed (Corollary 6.6, in its linearized fixed-input scope). Varying the cap across $\{1, 3, 6\}$ doubles as the sensitivity analysis of declared-budget results required in Section 6.

S7 Numerical validation of the fractional solver

This appendix collects numerical-method validation that supports the benchmark of Section 10 but is not part of its headline results.

The Caputo solver used throughout is the fractional Adams–Bashforth–Moulton PECE scheme of `benchmark/core.py`. Its convergence is validated against the exact Mittag–Leffler solution of the scalar test problem $D^\alpha y = \lambda y$, $y(0) = 1$, whose solution is $y(t) = E_\alpha(\lambda t^\alpha)$. Figure S1 shows the maximum absolute error as a function of the number of time steps on the benchmark grid; the error decreases under refinement and reaches 2.5×10^{-4} at the production resolution, as reported in Section 10.

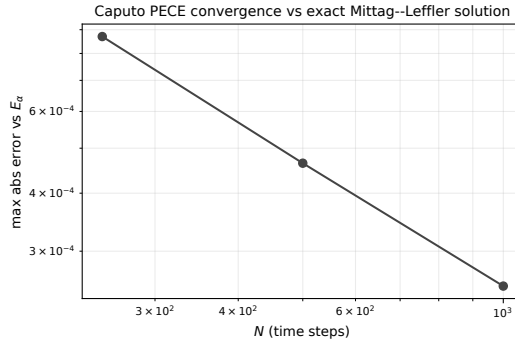


Figure S1: Caputo PECE solver convergence against the exact Mittag–Leffler solution: maximum absolute error versus the number of time steps, on a log–log scale. The monotone decrease under refinement is the numerical gate behind every nonlinear cell of Section 10.

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